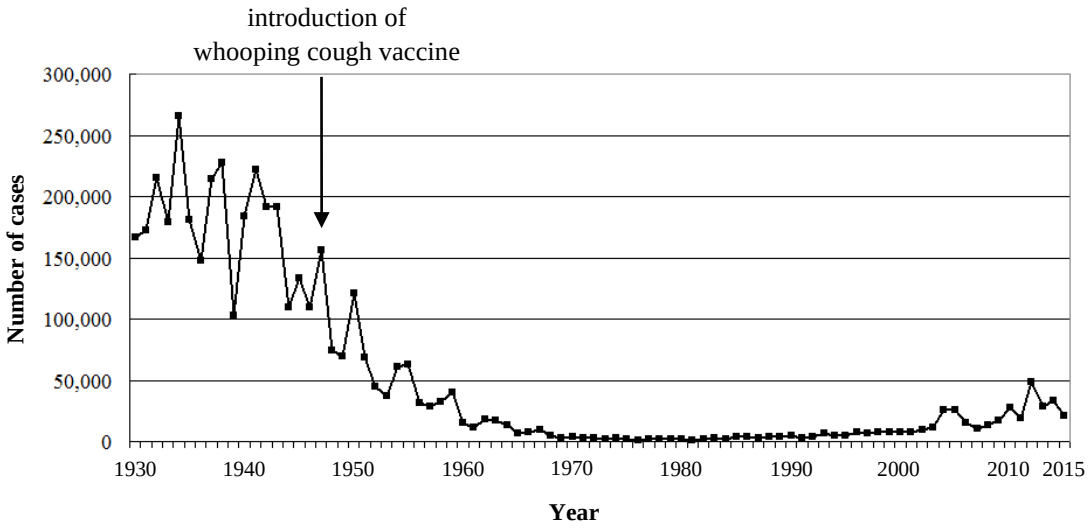
Short questions

|!|BCESC0250000001|!|

Whooping cough is a highly contagious bacterial disease. Prevention is mainly by vaccination. The graph below shows the number of reported cases in the United States from 1930 to 2015. The vaccine of whooping cough was introduced in the 1940s.



(a) State the type of immunity resulting from vaccination. (1 mark)

_____

_____

(b) Vaccination for whooping cough is recommended for children in the United States. With reference to the graph, describe the relationship between vaccination and the incidence of whooping cough. (2 marks)

_____

_____

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(c) Explain how vaccination helps prevent children from catching whooping cough.
(4 marks)

##

(a) Active immunity (1)

(b) Since the introduction of the whooping cough vaccine in the 1940s, the number of cases of whooping cough had been decreasing and remained at low levels. (1)
This shows that vaccination helps lower the incidence of whooping cough. (1)

(c) Vaccines contain antigens / pathogens (1)
which stimulate the immune system to produce memory cells for the particular antigens / pathogens. (1)
When the same antigens / pathogens invade the vaccinated person again, (1)
the memory cells produce large amounts of antibodies quickly to eliminate the invading pathogens. (1)

(7 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	1
(b)	Analysing data	1
	Applying concepts	1
(c)	Understanding basic concepts	4

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|!|BCESC0250000002|!|

Different parts of the skin act as physical barriers or chemical barriers against pathogens. Explain how the skin acts as

(a) a physical barrier, (2 marks)

(b) a chemical barrier. (2 marks)

##

(a) The outermost layer of the epidermis consists of dead cornified cells. (1)
It is waterproof and impermeable to pathogens. (1)

(b) Sebum is secreted by the sebaceous glands of the skin. (1)
It contains chemicals that kill certain pathogens. (1)

(4 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	2

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|!|BCESC0250000003|!|

When bacteria enter our bodies through a wound, phagocytes are attracted to the wound and they squeeze out of the capillaries.

(a) Describe what happens when phagocytes encounter bacteria. (4 marks)

(b) How is the process mentioned above facilitated by lymphocytes? (2 marks)

##

(a) Phagocytes form pseudopodia to surround the bacteria. (1)
The bacteria are enclosed in vacuoles. (1)
Lysosomes fuse with the vacuoles and release lysozyme into them. (1)
Lysozyme digests the bacteria. (1)

(b) Lymphocytes produce antibodies. (1)
Antibodies bind to the antigens of pathogens and help phagocytes detect pathogens more easily. (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	4
(b)	Understanding basic concepts	2

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|!|BCESB0250000004|!| ★

Susan carelessly cut her finger when she was working in the farm. The wound bled for a few minutes and then stopped.

(a) Describe the process that causes the bleeding to stop. (3 marks)

(b) Susan was later found to be infected with tetanus. Tetanus (破傷風) is caused by a bacterium which is generally found as spores in the soil and infects humans through wounds. The bacteria produce toxins that interfere with normal muscle contractions.
Describe how the injection of serum can provide treatment to Susan. (2 marks)

##

(a) Platelets are attracted to the wound. (1)
Platelets and the injured tissues release substances into the blood to convert fibrinogen into insoluble fibrin. (1)
Fibrins form a network which traps red blood cells, platelets, white blood cells and pathogens, forming a blood clot. (1)
The blood clot stops the bleeding.

(b) Serum contains a high concentration of ready-made antibodies. (1)
The injected antibodies act against the tetanus toxin and provide immediate immunity against tetanus. (1)

(5 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	3
(b)	Applying concepts	2

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|!|BCESC0250000005|!|

Explain how the inflammatory response leads to the following signs in the wound.

(a) Redness of the wound (2 marks)

(b) Swelling of the wound (3 marks)

(c) Pain (1 mark)

##

(a) Histamine released by mast cells causes the arterioles near the wound to dilate. (1)
This increases the blood flow to the wound and the wound becomes red. (1)

(b) Histamine released by mast cells increases the permeability of the capillary walls. (1)
More blood plasma and phagocytes are forced into the wound. (1)
Tissue fluid accumulates at the wound and the wound becomes swollen. (1)

(c) The swollen tissues press on and stimulate pain receptors in the skin. (1)

(6 marks)

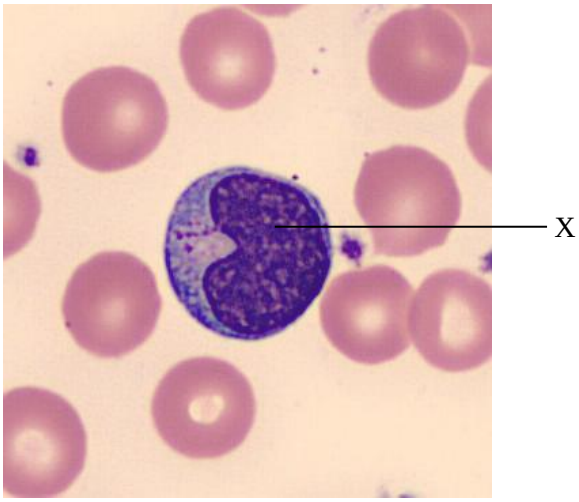
##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	3
(c)	Understanding basic concepts	1

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|!|BCESB0250000006|!| ★

The following photomicrograph shows some blood cells.



Source: https://commons.wikimedia.org/wiki/File:Hem1Lymphocyte.jpg

Credit: El*Falaf

(a) Name cell X and state its site of production. (2 marks)

(b) Cell X can differentiate into a cell which produces antibodies.

(i) Name the cell that can produce antibodies. (1 mark)

(ii) Describe ***two*** ways in which antibodies work against pathogens. (4 marks)

##

(a) lymphocyte (1)

Site of production: bone marrow (1)

(b) (i) plasma cell (1)

(ii) Any ***two sets*** of the following: (4)

- Some antibodies cause pathogens to clump together (1)
  thus preventing them from multiplying or entering body cells due to their large size. (1)

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- Some antibodies attach to the antigens of pathogens and make holes on the surfaces of pathogens. (1)
  The pathogens are lysed (burst open) and killed. (1)
- Some antibodies bind to the antigens of pathogens. (1)
  This helps phagocytes detect the pathogens more easily and hence facilitates phagocytosis (1)
- Some antibodies act as antitoxins by combining with the toxins secreted by pathogens and neutralize them. (1)
  This makes the pathogens harmless. (1)

(7 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
	Understanding basic concepts	1
(b)(i)	Understanding basic concepts	1
(b)(ii)	Understanding basic concepts	4

|!|BCESC0250000007|!|

For each of cell types listed in ***Column 1***, select from ***Column 2*** one description that matches it. Put the appropriate letter in the space provided. (4 marks)

Column 1		***Column 2***
plasma cell	(a) _____	A. multiplies and differentiate quickly during the secondary response
memory cell	(b) _____	B. destroys infected cells directly
killer T cell	(c) _____	C. produces antibodies
phagocyte	(d) _____	D. releases lysozyme
		E. releases histamine

##

(a) C (1)
(b) A (1)
(c) B (1)
(d) D (1)

(4 marks)

##

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	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1

|!|BCESC0250000008|!|

(a) Describe how vaccination can prevent influenza. (4 marks)

(b) Influenza has once been treated by serum injection in the United States in 1918 during an outbreak of influenza. List ***two*** differences between the types of immunity acquired by the injections of vaccine and serum. (2 marks)

##

(a) Vaccines contain weakened or killed influenza virus, or its antigen. (1)
After vaccination, antigens that have entered the human body stimulate the primary response and memory cells for the influenza virus antigen are produced. (1)
On subsequent exposure to the same antigen, the memory cells can initiate the secondary response quickly, leading to the production of a large amount of antibodies and killer T cells within a short time. (1)
The antibodies act against the influenza virus and the killer T cells destroy the cells infected with influenza virus. (1)

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(b) Any ***two*** of the following: (2)

Vaccination	Serum injection
Antibodies are produced by our own body.	Antibodies are transferred into the body.
It takes time to develop immunity and thus cannot provide immediate immunity.	It provides immediate immunity.
Duration of protection is long.	Duration of protection is short.

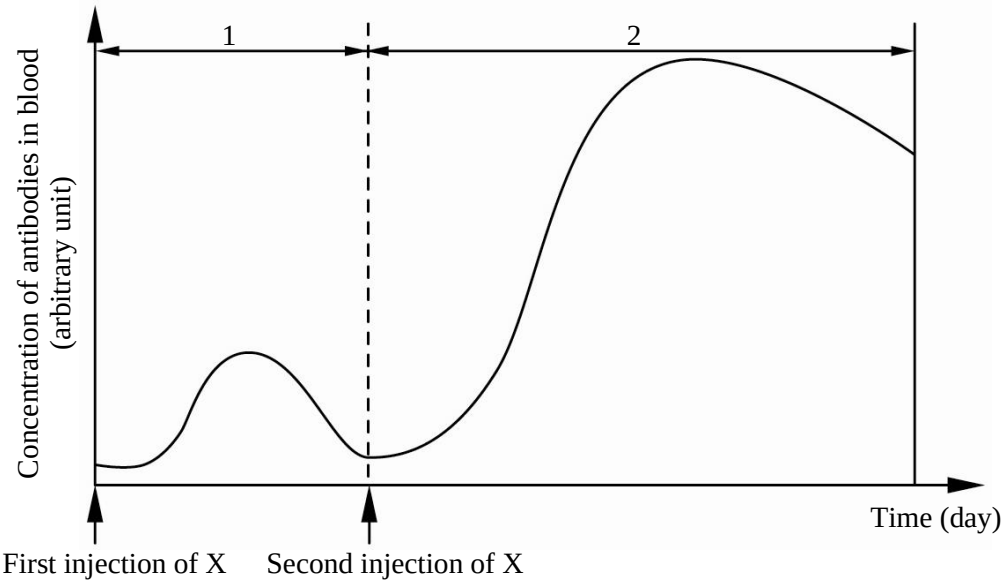
(6 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	4
(b)	Applying concepts	2

|!|BCESB0250000009|!| ★

In an investigation, substance X was injected into a human body at two different times and the changes in the concentration of antibodies in the blood was recorded. The following graph shows the results.



(a) Suggest what substance X may be. (1 mark)

(b) Based on the graph, state ***two*** differences between the curves in periods 1 and 2. (2 marks)

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(c) Describe ***one*** abnormality shown in the graph. Explain your answer. (2 marks)

(d) What is the type of immunity shown in the above graph? (1 mark)

##

(a) antigen / pathogen / vaccine (1)

(b)

	Period 1	***Period 2***	
Maximum concentration of antibodies in blood	lower	higher	(1)
Decline in concentration of antibodies in blood	quicker	slower	(1)

(Or other reasonable answers)

(c) During period 2, the latent period should be shorter / the concentration of antibodies in blood should rise to the maximum level within a shorter period of time. (1)

This is because the memory cells formed after the first infection can initiate faster secondary response / produce antibodies within a short time upon exposure to the same antigens. (1)

(d) active immunity (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Analysing data	2
(c)	Analysing data	1
	Applying concepts	1
(d)	Analysing data	2

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|!|BCESC0250000010|!|

Fill in the blanks to complete the following paragraph. (5 marks)

(a)_____ immunity is acquired naturally after recovering from a disease, or can be acquired (b)_____ through vaccination. After vaccination, antigens from the vaccine stimulate the primary response and (c)_____ for the particular antigen are produced. On subsequent exposure to the same antigen, the (d)_____ response is initiated quickly, leading to the production of a large amount of antibodies and (e)_____ within a short time. This provides immunity to the body.

##

(a) Active (1)
(b) artificially (1)
(c) memory cells (1)
(d) secondary (1)
(e) killer T cells (1)

(5 marks)

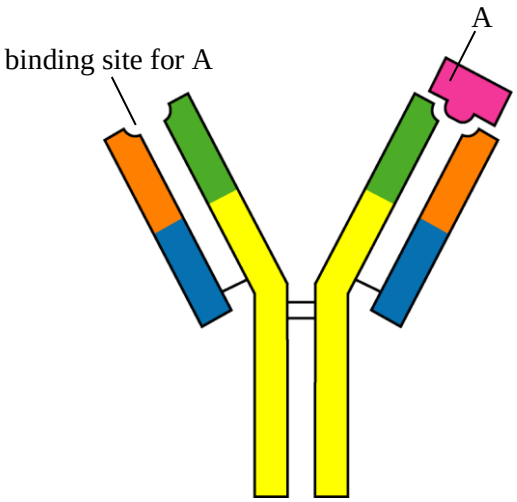
##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1

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|!|BCESB0250000011|!| ★

The following diagram shows the structure of a Y-shape antibody molecule.



(a) Identify substance A. (1 mark)

(b) State the cell that produces antibodies. Describe ***two*** adaptive features of this cell for antibody production. (3 marks)

(c) State ***two*** actions of antibodies against pathogens. (2 marks)

##

(a) antigen (1)

(b) Antibody-producing cells are plasma cells. (1)

Any ***two*** of the following adaptive features: (2)

- abundance of rough endoplasmic reticulum
- abundance of mitochondria
- abundance of ribosome

(Or other reasonable answers)

(c) Any ***two*** of the following: (2)

- make pathogens clump together
- make holes on the surfaces of pathogens, leading to lysis of pathogens
- enhance phagocytosis that destroys pathogens

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- combine with the toxins secreted by pathogens and neutralize them, making them harmless

(6 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Understanding basic concepts	3
(c)	Understanding basic concepts	2

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